(a)(i)

$$f(12) = 12\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right) = \boxed{4}$$

$$f(180) = 180\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{5}\right) = \boxed{48}$$

(a)(ii) Let $N = p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_k^{\alpha_k}$, where $p_1, \ldots, p_k$ are distinct primes and $\alpha_1, \ldots, \alpha_k$ are positive integers.

$$\begin{aligned} f(N) &= p_1^{\alpha_1}\cdots p_k^{\alpha_k}\left(1-\frac{1}{p_1}\right)\cdots\left(1-\frac{1}{p_k}\right) \\ &= p_1^{\alpha_1-1}(p_1-1)\cdots p_k^{\alpha_k-1}(p_k-1) \end{aligned}$$

Thus $f(N) \in \mathbb{Z}$

(b)(i) $\boxed{\text{False}}$

$$f(3) = 2, \quad f(9) = 6, \quad f(27) = 18$$
$$f(3)f(9) = 12 \neq 18 = f(27)$$

(b)(ii) $\boxed{\text{True}}$

$$\begin{aligned} f(p)f(q) &= p\left(1-\frac{1}{p}\right)q\left(1-\frac{1}{q}\right) \\ &= pq\left(1-\frac{1}{p}\right)\left(1-\frac{1}{q}\right) \\ &= f(pq) \end{aligned}$$

(b)(iii) $\boxed{\text{False}}$

$$f(1) = 1, \qquad f(3) = 2$$
$$f(1)f(3) = 2 = f(3) = f(1\cdot 3)$$

But 1 is not prime

(c) For prime p

$$\begin{aligned} f(p^m) &= p^m\left(1-\frac{1}{p}\right) \\ &= p^{m-1}(p-1) \end{aligned}$$

$$\because 146410 = 10 \cdot 11^4$$

$$\therefore p - 1 = 10, \qquad p^{m-1} = 11^4$$

$$\boxed{p = 11, \qquad m = 5}$$

This solution was submitted by **Jon**.

Published and refined by **Jon**.

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